WIEBE'S THEOREM RELATIVE TO AN IDEAL, AND ITS UNIT FORMALISATION

This note presents a variation of the proof of Wiebe's theorem in T. Coquand and C. Tête, *An elementary proof of Wiebe's Theorem* (J. Algebra 499, 2018; `ctete.pdf`, Theorem 11 and Corollary 12), and its checked formalisation in Unit (`sag/wiebe`). The argument is the one of the paper. Three points are made explicit: everything is stated relative to a finitely generated ideal J instead of in quotient rings; the generic change of the sequences is a finite chain of elementary operations, each of which visibly keeps the determinant; and the rings obtained by adding indeterminates enter only through two properties of the inclusion maps.

1. Statement

Let R be a commutative ring and J a finitely generated ideal. For a list $c = (c_1, \dots, c_n)$ write (J, c) for the ideal generated by J and the c_i. An element r is *regular mod* J if $rx \in J$ implies $x \in J$; a list is regular mod J if $c_i x \in J$ for all i implies $x \in J$; $c_1, \dots, c_k$ is a regular sequence mod J if each c_i is regular mod $(J, c_1, \dots, c_{i-1})$. The true grade $\mathrm{Gr}(I) \geq k$ mod J of a finitely generated ideal I (ctete, Definition 5, relative to J) means that, merely, after adding finitely many indeterminates, I contains a regular sequence of length k mod J. Theorem 9 of ctete holds in this form: if $\mathrm{Gr}(I) \geq k+1$ mod J and $r \in I$ is regular mod J, then $\mathrm{Gr}(I) \geq k$ mod (J, r).

Theorem 1. *Let $a, b \in R^n$ and let V be an $n \times n$ matrix with $b_j \equiv \sum_i a_i V_{ij}$ mod J for all j, and $\Delta = \det V$.*

1. *For every i, $\Delta a_i \in (J, b)$. Hence $x \in (J, b, a)$ implies $x\Delta \in (J, b)$, and $x \in (J, b, \Delta)$ implies $xa_i \in (J, b)$ for all i.*
2. *If $\mathrm{Gr}(b) \geq n$ mod J, the converses hold: $x\Delta \in (J, b)$ implies $x \in (J, b, a)$, and $xa_i \in (J, b)$ for all i implies $x \in (J, b, \Delta)$.*

So, modulo (J, b), $\Delta^\perp = (a)$ and $(a)^\perp = (\Delta)$.

With $J = 0$ this is ctete Theorem 11. Since a regular sequence b mod J of length n gives $\mathrm{Gr}(b) \geq n$ mod J directly, it contains Corollary 12.

Part 1 is Cramer's rule: writing $b \equiv aV$ for row vectors, $b\,\mathrm{adj}(V) \equiv aV\mathrm{adj}(V) = \Delta a$ mod J, so $\Delta a_k \equiv \sum_j b_j \mathrm{adj}(V)_{jk} \in (b)$. The rest of this note proves part 2, by induction on n, for all rings R and ideals J at once. For $n = 0$, $\Delta = 1$ and both statements are trivial.

2. Three invariance properties

Elementary operations. For an index r and coefficients w with $w_r = 0$, the elementary operation $c \mapsto c'$ adds $\sum_l w_l c_l$ to c_r and leaves the other entries. It is undone by the operation with $-w$, since $\sum_l w_l c'_l = \sum_l w_l c_l$ (because $w_r = 0$), so $(J, c') = (J, c)$. Its matrix E has determinant 1: up to transposition it is the identity with one column replaced by $e_r + w$, and $\det I[r \leftarrow v] = v_r$ because $v \mapsto \det I[r \leftarrow v]$ is linear and takes the value δ_{kr} at e_k.

- An operation on b replaces V by VE (in the row-vector convention), so $b' \equiv aV'$ mod J with $\det V' = \Delta$.
- An operation on a gives $b \equiv a'E^{-1}V$, with $\det(E^{-1}V) = \Delta$, and $(a') = (a)$.

Date: October 25, 2026.

The conclusions of part 2 only depend on the ideals (J,a), (J,b) and on Δ: if $(J,b')=(J,b)$ and every a'_i is in (J,a), the conclusions for (a',b',Δ) give those for (a,b,Δ). So part 2 for (a,b,V) follows from part 2 for any data reached from it by elementary operations.

Good ring maps. Call $h:R\to T$ *good* if it keeps regularity mod lists (r regular mod L implies $h(r)$ regular mod $h(L)$) and has a retraction $\rho:T\to R$ (a ring map with $\rho h=\mathrm{id}$). The inclusion $R\to R[X]$ is good (McCoy for the first property, evaluation at 0 for the second), and composites of good maps are good. Along any ring map, $b\equiv aV \bmod J$ becomes $h(b)\equiv h(a)h(V) \bmod h(J)$ and $\det h(V)=h(\Delta)$; along a good map, the conclusions of part 2 for $h(J),h(a),h(b),h(\Delta)$ give those for J,a,b,Δ, since ρ brings memberships of images back.

Changing the ideal. If $\mathrm{Gr}(I)\geq k \bmod J$ and every element of I is in (J,I'), then $\mathrm{Gr}(I')\geq k \bmod J$: each term of a regular sequence in I is $t+j$ with $t\in(I')$, $j\in J$, and a regular sequence mod J stays one when its terms are changed mod J.

3. The generic position

Lemma 2. *Let $\mathrm{Gr}(b_1,\dots,b_n)\geq n$ mod J. Then, merely, along a good map $h:R\to T$, the list $h(b)$ is changed by elementary operations, not touching b_n, into b' such that $b'_1,\dots,b'_{n-1},h(b_n)$ is a regular sequence mod $h(J)$.*

Proof. Induction on n. For $n=1$, $\mathrm{Gr}(b_1)\geq 1$ means that b_1 is regular mod J; take $h=\mathrm{id}$. For $n+1$, in $R[u]$ let

$$f=b_1+u\,(b_2+u\,b_3+\cdots+u^{n-1}b_{n+1}),$$

the Kronecker polynomial of $(b_1,\dots,b_{n+1})$ with constant term b_1. Since (b) is regular mod J ($\mathrm{Gr}\geq 1$), f is regular mod J (McCoy, ctete Theorem 4), and $f\in(b)$. After the base change to $R[u]$, Theorem 9 gives $\mathrm{Gr}(b)\geq n$ mod (J,f). As $b_1=f-u(\dots)$ is in $(J,f,b_2,\dots,b_{n+1})$, $\mathrm{Gr}(b_2,\dots,b_{n+1})\geq n$ mod (J,f). By induction, over $R[u]$ and mod (J,f), a good map h' and operations give $b'_2,\dots,b'_n,h'(b_{n+1})$ regular mod (J,f). Replacing b_1 by f is one elementary operation ($f-b_1=\sum_{l\geq 2}u^{l-1}b_l$), and the operations on $b_2,\dots,b_{n+1}$ are operations on the whole list (with coefficient 0 at the first place). So $h=h'\circ(R\to R[u])$ and $b'_1=h'(f)$ give a regular sequence $b'_1,\dots,b'_n,h(b_{n+1})$ mod $h(J)$. □

The polynomials b'_i are ctete's $b_i+u_ib_{i+1}+u_i^2b_{i+2}+\cdots$; here they arise one at a time, each as one elementary operation.

4. The induction step

Assume part 2 for n, for all rings and ideals, and let $a,b\in R^{n+1}$, V, $\mathrm{Gr}(b)\geq n+1$ mod J. By Lemma 2 and the invariance properties we may assume (after a good map, renaming $h(J)$ as J) that

$$b_1,\dots,b_n,b_{n+1}\text{ is a regular sequence mod }J.$$

Write $K=(J,b_1,\dots,b_n)$; so b_{n+1} is regular mod K.

Position of a. If x kills (a) mod an ideal containing J, it kills (b), since $b\equiv aV$. Hence (a) is regular mod J (as (b) is) and mod K (as b_{n+1} is). In $R[u]$ let

$$a'_{n+1}=a_{n+1}+u\,(a_n+u\,a_{n-1}+\cdots+u^{n-1}a_1).$$

It is regular mod J and mod K (McCoy), and replacing a_{n+1} by a'_{n+1} is an elementary operation on a. We rename a' as a.

The grade mod a_{n+1}. Replacing the last term of a regular sequence by an element regular modulo the previous terms gives a regular sequence: so $b_1, \dots, b_n, a_{n+1}$ is a regular sequence mod J, in the ideal $(b_1, \dots, b_n, a_{n+1})$, which therefore has $\mathrm{Gr} \geq n+1$ mod J. By Theorem 9 with a_{n+1}, it has $\mathrm{Gr} \geq n$ mod (J, a_{n+1}), hence, as $a_{n+1} \in (J, a_{n+1})$,

$$\mathrm{Gr}(b_1, \dots, b_n) \geq n \text{ mod } J' = (J, a_{n+1}).$$

Modulo J', $b_j \equiv \sum_{i \leq n} a_i V_{ij}$ for $j \leq n$, that is, $b_< \equiv a_< W$ with W the upper left $n \times n$ block of V; let $\delta = \det W$. By induction, mod $(J', b_<) = (K, a_{n+1})$: $x\delta \in (K, a_{n+1})$ implies $x \in (K, a_{n+1}, a_<)$, and $ya_i \in (K, a_{n+1})$ for all $i \leq n$ implies $y \in (K, a_{n+1}, \delta)$.
Cramer. By part 1, $\Delta a_{n+1} \equiv \sum_j b_j \mathrm{adj}(V)_{j,n+1}$ mod J, and $\mathrm{adj}(V)_{n+1,n+1} = \delta$; hence

$$\Delta a_{n+1} \equiv \delta\, b_{n+1} \pmod{K}.$$

First inclusion. Let $x\Delta \equiv b_{n+1} y$ mod K. Then $b_{n+1} x\delta = x\delta b_{n+1} \equiv x\Delta a_{n+1} \equiv b_{n+1} a_{n+1} y$ mod K, so $x\delta \equiv a_{n+1} y$ mod K as b_{n+1} is regular mod K. Thus $x\delta \in (K, a_{n+1})$, and by induction $x \in (K, a_{n+1}, a_<) \subseteq (J, b, a)$.
Second inclusion. Let $xa_{n+1} \equiv b_{n+1} y$ and $xa_i \equiv b_{n+1} y_i$ mod K for $i \leq n$. Then

$$b_{n+1} y a_i \equiv a_i x a_{n+1} = a_{n+1} x a_i \equiv b_{n+1} a_{n+1} y_i \pmod{K},$$

so $ya_i \equiv a_{n+1} y_i$, i.e. $ya_i \in (K, a_{n+1})$, for all $i \leq n$. By induction $y \equiv a_{n+1} t + \delta s$ mod K. Then

$$a_{n+1} x \equiv b_{n+1} y \equiv a_{n+1} b_{n+1} t + \delta b_{n+1} s \equiv a_{n+1}(b_{n+1} t + \Delta s) \pmod{K},$$

and since a_{n+1} is regular mod K, $x \equiv b_{n+1} t + \Delta s$ mod K: $x \in (J, b, \Delta)$.
Back to the data. The conclusions hold for the changed data in the extended ring; by the invariance properties they hold for the images of a, b, Δ (elementary operations keep (J, b), put a' in (J, a) and keep Δ), and then for a, b, Δ in R (good map). This ends the induction and the proof of Theorem 1.

Remark 3. Compared with ctete: the statement is relative to J and the ring $T = R/(a_n)$ becomes the ideal (J, a_{n+1}), which is what makes the induction go through without quotient rings; the claim $\mathrm{Gr}(b_1, \dots, b_n) \geq n$ mod (J, a_{n+1}) is derived explicitly by replacing the last term of the generic regular sequence by a'_{n+1} before applying Theorem 9; and "without changing Δ" is the fact that each elementary matrix has determinant 1.

5. The formalisation

The proof is checked in `sag/wiebe` (168 modules with the linked foundation, about 2.6 s; each of the eight files below checks in under 0.1 s, and no declaration opens more than 226 definitions). It uses the true grade and Theorem 9 of `sag/koszul`, McCoy modulo an ideal from `sag/cffr`, and determinants, the adjugate, transposition and the naturality of det from `sag/exterior`. Nothing is assumed.

Matrices are lists of columns, and the hypothesis is $b_j \equiv \sum_i a_i V\, i\, j$; the columns of the formal V are the rows of the V above, which makes Cramer's rule the identity $\mathrm{adj}(V)V = \det V \cdot I$ without transposition. The lists (J, c) are built with the last generator outermost, so that $x \in (L, c)$ iff $x - cw \in L$ for some w.

`WiebeLists`	the lists (J, c), inclusion by generators, congruence mod an ideal
`WiebeMatrices`	elementary operations and $\det E = 1$; Cramer at the last index; $\mathrm{adj}(V)_{nn} = \det W$; the block hypothesis
`WiebeCore`	the two inclusions at $n+1$ from n (`Step`), and $n = 0$
`WiebeSequences`	last term of a regular sequence, replacing it, changing the ideal; good maps
`WiebeChain`	chains of elementary operations: ideals, V and det, along ring maps
`WiebeGeneric`	Lemma 2 (`generic`)
`WiebeTransfer`	invariance of the conclusions; descent along good maps
`Wiebe`	the induction (`theorem11`); the equalities (`Thm11.WithGrade`); Corollary 12 (`Thm11.WithSequence`)

Lemma 2 quantifies over the extension ring; it is stated as "for every proposition P, if every such extension gives P, then P", so that no type of rings has to be truncated.